

Charlie- BE Code Assessment

Senior Backend Software Engineer

Introduction

Thank you for your interest in the Senior Backend Software Engineer role at Charlie. As part of our evaluation process, we would like you to complete a take-home code assessment. This assessment will evaluate your skills in design, testing and development using Java and Spring Boot.

If you have any questions or need clarification during the assessment, feel free to reach out to the recruiter. Good luck with your assessment, and we look forward to reviewing your work!

Task

Your task is to design a parking lot service using object-oriented principles. The [starter project \(parking-lot-service.zip\)](#) includes an in memory db and a few other dependencies. Please spend no more than four hours. This ensures fairness and consistency in the evaluation.

Assumptions

1. The parking lot has 25 total spaces
2. The parking lot can hold motorcycles, cars and vans
3. The parking lot has motorcycle spots, compact car spots and regular spots
4. A car can park in a single compact spot, or a regular spot
5. A van can park, but it will take up 3 regular spots (for simplicity we don't need to make sure these spots are beside each other)

Acceptance Criteria

1. Service endpoints
 - a. Park vehicle
 - b. Vehicle leaves parking lot
 - c. Find how many spots are remaining
 - d. Check if all parking spots are taken for a given vehicle type

Submission

Please submit your completed assessment by emailing a zip file containing your project files to the recruiter or by sharing the GitHub repository link if you choose to use GitHub.